KAPASA MAKASA UNIVERSITY

School of Applied Sciences and Open Learning | ICT Department

Practical Test 2 — Advanced Integrated Assessment

Hello, and welcome to your Data Visualization practical. This paper is designed to genuinely stretch you — it asks you to make judgement calls the way a real analyst would, not just follow a checklist. Read each scenario carefully, work methodically, and don't panic if a step takes longer than expected: partial, well-reasoned progress is valued throughout the marking scheme. Take a breath, manage your time, and enjoy the challenge — we're rooting for you!

Course Name: Data Visualization with Excel, Power BI and Tableau

Duration: 2 Hours

Course Code: DS 231

Total Marks: 100

EXAMINATION INSTRUCTIONS

- **Format:** This is a strictly practical, hands-on examination spanning **three tools** — Microsoft Excel, Microsoft Power BI, and Tableau Desktop. You are required to demonstrate technical proficiency and independent analytical judgement across all three.
- **Resources:** You may only use the provided dataset(s) and the software installed on your workstation. No AI assistants, templates, or pre-built workbooks may be used.
- **Time Management:** You are not required to attempt the questions in order, but a suggested pacing of approximately 35–40 minutes per tool is recommended given the volume of work.
- **Submission:** At the end of the examination, you must submit **three files** to the following Google Drive link:

  https://drive.google.com/drive/folders/2gT4mPz9RxKbAqLwN6vYsHc8jFdEo3ZtB

  Ensure your files are named:
  - [YourStudentID]_Q1.xlsx
  - [YourStudentID]_Q2.pbix
  - [YourStudentID]_Q3.twbx
- **Integrity:** Independent work is required. Any detected file sharing, duplication, or identical calculated-field logic across submissions will result in an automatic zero for all parties involved.
- **Save Frequently:** It is your responsibility to save your work continuously in all three applications.

Dataset Requirements

Dataset Title: Online Retail II — UK-Based Online Retailer (December 2009 – December 2011)

Download Link (choose either mirror):

https://archive.ics.uci.edu/dataset/502/online+retail+ii

https://www.kaggle.com/datasets/mathchi/online-retail-ii-data-set-from-ml-repository

Context: This dataset contains genuine, uncleaned transactional records from a UK-based online retailer of unique all-occasion giftware, spanning two full trading years across two internal extracts ("Year 2009-2010" and "Year 2010-2011"). It is substantially messier than a typical classroom dataset and includes, among other issues:

- Cancelled transactions, identifiable only because their **InvoiceNo** begins with the letter "C" — these must be reasoned about programmatically, not filtered by eye.
- Negative and zero values in **Quantity** and **Price**, representing returns, damages, samples, and manual price adjustments.
- Non-merchandise **StockCode** values mixed in with real products — e.g. *POST, DOT, M, BANK CHARGES, AMAZONFEE, CRUK, gift card codes* — which will silently distort revenue totals if not identified and excluded.
- Approximately 20% of rows with a missing/blank **Customer ID**, which cannot simply be dropped without materially understating revenue.
- Inconsistent country naming — e.g. *"EIRE"* instead of Ireland, *"Unspecified"*, *"European Community"* — requiring standardisation before any geographic aggregation is trustworthy.
- Exact duplicate rows, inconsistent casing and stray whitespace in **Description**, and **InvoiceDate** values that some export tools store as text rather than a true datetime.

Candidates are expected to make and justify their own reasonable data-cleaning decisions where the instructions do not spell out an exact rule — this is treated as part of the analytical skill being assessed, not an ambiguity to be resolved by guessing at a single "correct" answer.

Tips for Success

- Skim all three questions first — some cleaning logic (cancellations, non-merchandise codes, country names) is reused across Excel, Power BI, and Tableau, so decide your rules once and apply them consistently.
- Build your helper/flag columns before your visuals — a dashboard built on unreliable numbers will lose marks even if it looks polished.
- If you get stuck on one sub-task, move on and come back — every question is marked in discrete parts, so partial credit is always available.
- Name your worksheets, pages, and dashboards clearly — a marker who can navigate your file quickly is a marker who can find your best work.

Question 1: Advanced Data Transformation & Interactive Dashboard 30 Marks

Software: Microsoft Excel (Power Query + PivotTables)

Scenario: You are the lead Data Analyst for the retailer. Finance has asked for a historically accurate, formula-driven revenue reconstruction ahead of a two-year performance review — manual filtering or eyeballing rows to remove "bad" data is explicitly not acceptable, since the workbook must remain auditable and reproducible if the raw export changes next quarter. The Head of Analytics has also asked for a genuine data-science lens on the same data: which customers actually matter, and which products are quietly carrying the business.

Tasks:

1. **Formula-Driven Cleaning Flags:** Import and combine both yearly extracts into one working table. Without deleting any raw rows, add calculated helper columns that use formulas (e.g. *LEFT, ISNUMBER, MATCH/XLOOKUP, TRIM, EXACT*) to flag: (a) cancelled invoices, (b) non-merchandise stock codes (using a small list you construct on a separate sheet), and (c) exact duplicate line items.
2. **Calculated Columns & Outlier Detection:** Create **Line Value** (Quantity × Price, correctly signed so returns appear negative) and an **Order Type** column (Sale / Return / Cancelled / Non-Merchandise) using nested *IF* or *IFS* logic. Then

add an **Outlier Flag** column that uses a formula-based interquartile-range (IQR) rule — *QUARTILE.INC* and a 1.5×IQR fence — to flag unusually large line values that could distort your totals if left unexamined.

3. **PivotTable-Driven RFM & Pareto Analysis:** On a new worksheet, build a customer-level **RFM** table using PivotTables and formulas: *Recency* (days since each customer's last purchase, relative to the dataset's latest date), *Frequency* (distinct invoice count per customer), and *Monetary* (Net Revenue per customer) — excluding cancelled and non-merchandise rows throughout. Use these to assign each customer a **Customer Segment** (High / Medium / Low value) with a formula. Separately, build a **Pareto (80/20) analysis** of products: a PivotTable of Net Revenue by product with a running-total % column, identifying what share of products generates 80% of total revenue.
4. **Executive Dashboard:** On a dedicated "Executive Dashboard" worksheet, using PivotCharts connected to your cleaned data, build: a line chart of monthly Net Revenue across both years; a bar chart of the Top 10 customers by Net Revenue; a chart showing the distribution of customers across your RFM segments; a slicer for Year and Customer Segment connected across the charts; and conditional formatting that highlights any outlier line items still present in the underlying data view.

Marks Allocation: Formula-Driven Cleaning Flags (7), Calculated Columns & Outlier Detection (7), RFM & Pareto Pivot Analysis (9), Dashboard Design & Slicer Interactivity (7).

Stretch Challenge (optional, no extra marks — but excellent practice)

Add a dynamic named range or Excel Table so your PivotTables auto-expand if next quarter's export has more rows, and see whether your Pareto analysis still roughly holds to an 80/20 split once outliers are excluded.

Question 2: Enterprise BI Modelling with Time Intelligence 35 Marks

Software: Microsoft Power BI

Scenario: The Business Intelligence team wants a repeatable, governed model that survives a raw re-export of the data, and that can explain not only revenue but how it is trending year over year and where returns are concentrated. You are also given a small supplementary **Country–Region reference table** (with a few of its own inconsistent country spellings) that must be merged in rather than typed in by hand.

Tasks:

5. **Power Query Transformations:** Import both yearly extracts and append them into a single query. Promote headers; correct the **InvoiceDate** data type; add a conditional column **IsCancelled** (InvoiceNo starts with "C"); filter out rows whose **StockCode** matches a supplied non-merchandise list; and instead of deleting blank Customer IDs outright, replace them with a flag value (e.g. "Guest") so guest-checkout revenue is retained but reportable separately. Merge the Country–Region reference table onto the main query and resolve any spelling mismatches so no country is left unmatched (null Region).
6. **DAX Foundations:** Create explicit measures: **Net Revenue** (SUMX of Quantity × Price, excluding cancelled rows), **Return Value**, **Total Orders** (distinct count of InvoiceNo, excluding cancelled), **Average Order Value**, **Return Rate %**, and a time-intelligence measure **Revenue YoY Growth %** using *SAMEPERIODLASTYEAR* or *DATEADD*.
7. **Core Visualisations & Drillthrough:** On Page 1, build Card visuals for Net Revenue, Return Rate %, and Revenue YoY Growth %; a line chart of monthly Net Revenue with the prior-year trend overlaid for comparison; and a clustered bar chart of the Top 10 countries by Net Revenue with the United Kingdom excluded via a visual-level filter, to focus on international performance. Add a drillthrough page that shows customer-level order detail when a country bar is right-clicked.

Marks Allocation: Power Query Transformations & Reference Merge (13), DAX Measure & Time-Intelligence Accuracy (13), Visualisation, Filtering & Drillthrough (9).

Stretch Challenge (optional, no extra marks — but excellent practice)

Add a measure that flags months where Return Rate % worsened versus the prior year, and format it conditionally on your line chart.

Question 3: Integrated Customer Analytics & Storytelling 35 Marks

Software: Tableau Desktop

Scenario: The CEO wants a customer-centric story for the same two-year period: which customers actually drive the business, how concentrated that value is, and where the international opportunity lies. You must build this directly on top of the raw two-year extracts using the Level of Detail and parameter techniques from the lab programme, not on a pre-cleaned copy.

Tasks:

8. **Data Modelling:** Connect to both yearly extracts and combine them with a **Union** (not a join) into a single logical table. Create a calculated field **Order Status** (Sale / Return / Cancelled) and a **Net Revenue** calculation that correctly signs returns, mirroring the logic you used in Excel.
9. **LOD Expressions, Sets & Parameters:** Build a **FIXED** LOD field **Customer Lifetime Value** (total Net Revenue per Customer ID across the full two-year window, independent of any filters in the view). Create an integer parameter **Top N Selector** and use it to drive a dynamic Top-N **Set** of customers by Lifetime Value. Using the FIXED LOD result, build a **Customer Segment** field bucketing customers into High / Medium / Low tiers.
10. **Executive Dashboard:** On a new dashboard, include: a filled map of Net Revenue by country (non-merchandise stock codes excluded); a bar chart of the parameter-controlled Top-N customers by Lifetime Value; a donut or treemap of Revenue share by Customer Segment; and a date-range filter covering the full two-year period. Add a dashboard filter action so that clicking a country on the map filters the customer bar chart to that country's customers only.
11. **Data Storytelling & Polish:** Insert a text box directly onto the dashboard and write two distinct, actionable insights based entirely on the visuals you built (for example, about revenue concentration or an international opportunity). Ensure a consistent colour theme, clear titles, and appropriate data labels throughout.

Marks Allocation: Data Modelling & Union Logic (6), LOD/Parameter/Set Construction (12), Dashboard Layout & Interactivity (11), Analytical Insights & Polish (6).

Stretch Challenge (optional, no extra marks — but excellent practice)

Add a trend line or forecast to your Top-N customer chart, and see if your two insights still hold once you account for it.

That's the full paper. Take one more look at your file names and submission link before you finish up — well done for working through this, and good luck!

— End of Practical Test 2 —